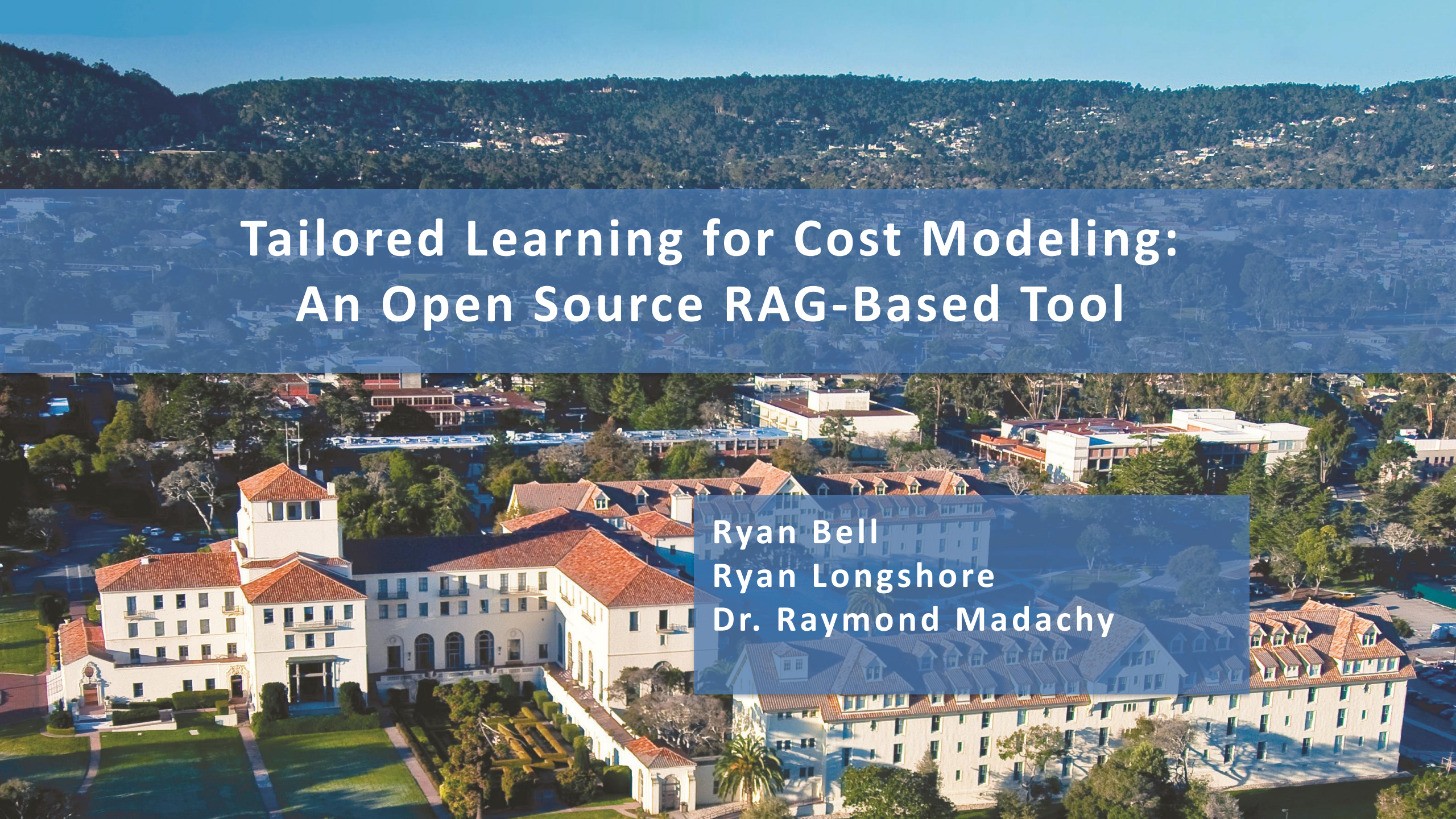




Tailored Learning for Cost Modeling: An Open Source RAG-Based Tool

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Agenda

- Abstract
- Large Language Model (LLM) Overview
- Improving LLM Responses
- Building the RAG Pipeline
- Creating the GUI
- Demonstration



Abstract

For professionals entering the specialized field of cost modeling, there is often a **significant gap in domain-specific knowledge**. To address this, we present an AI-driven approach that leverages **Python's Haystack** package to employ a **Retrieval-Augmented Generation (RAG) pipeline** to offer customized interactions with foundational cost modeling resources, including **Barry Boehm's books and other prominent cost modeling documents**. This presentation explores the integration of this AI tool to deliver on-demand, conversational access to cost modeling knowledge, as well as a secondary feature for generating multiple-choice questions to reinforce learning.

The RAG pipeline enables engineers to **query specific documents and receive relevant information from established cost modeling tools**, simulating a mentor-mentee interaction and providing a much-needed knowledge repository. By dynamically generating both responses to complex inquiries and formative assessment items, this system aims to facilitate the understanding and application of core cost modeling principles for those new to the field. **Initial deployment results will be presented**, demonstrating its potential to fill knowledge gaps and improve the ability of engineers to contribute effectively to cost estimation projects.



Large Language Model (LLM) Overview

- Large Language Models (LLMs): Neural networks that have transformed natural language procession (NLP) by demonstrating capability to generate relevant text based on a human user's prompt.
- Common LLMs
 - Proprietary: Anthropic Claude, OpenAI ChatGPT, Google Gemini
 - Open Source: Mistral, LLaMa, Falcon
- Characterizing Models
 - Size is the number of weights, also known as parameters
 - Quantization converts weights to lower precision data types to reduce memory and compute requirements
- Challenges
 - **Hallucinations**
 - **Access to Unique Training Data**



There are other challenges, but these are the two most prominent for this presentation

Improving LLM Responses

Training:

The process of teaching an LLM to perform tasks by exposing it to large datasets and optimizing it to recognize patterns, relationships, and context within the data.

Inferencing:

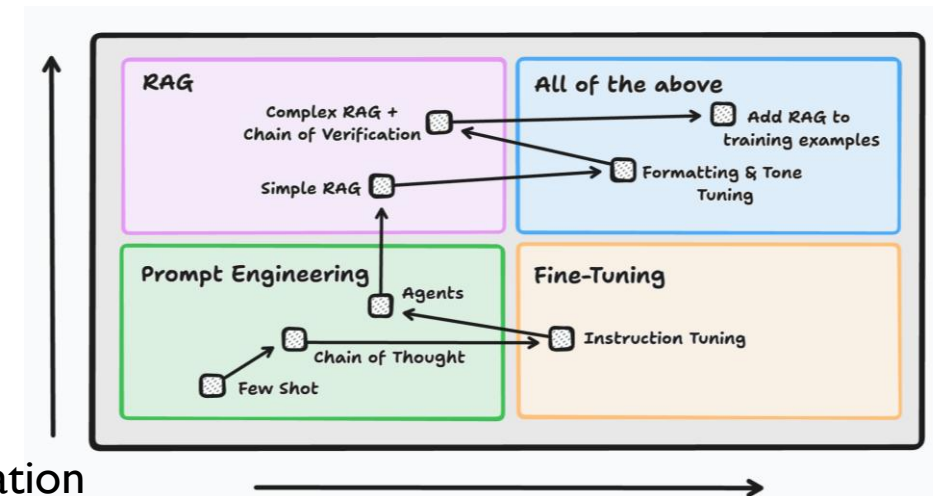
The LLM's process of generating responses or predictions based on new inputs, using the knowledge it gained during training.

- **Re-Training:** Re-train base model with new dataset
- **Finetuning:** Teach the base model specialized knowledge by adjusting a small number of internal parameters. Significant time savings compared to re-training entire model
- **Retrieval Augmented Generation (RAG):** Utilize external data to augment the LLM's knowledge without changing model weights
- **Prompt Engineering:** Optimize prompts for optimal outputs
 - **Few Shot** prompts teach LLMs to perform a task through examples
 - **Zero Shot** prompts challenge the LLM to perform a task correctly when it has not been correctly trained for that task

Context

Optimization

What the model needs to know



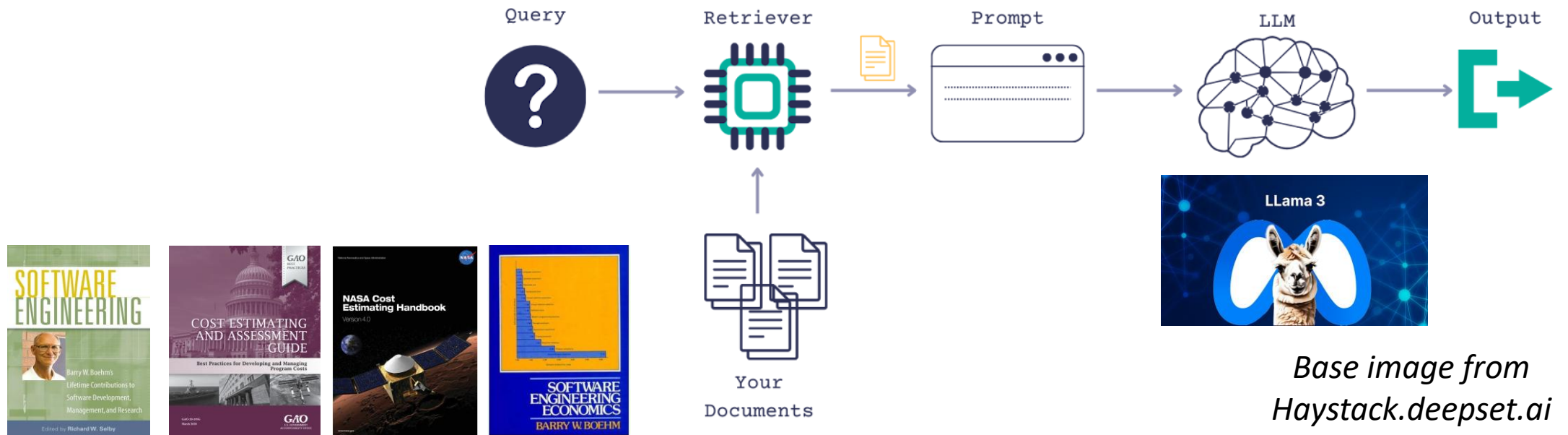
LLM Optimization

How the model needs to act

COMPUTE / COMPLEXITY / COST

Building the RAG Pipeline

- The Haystack Python package was utilized to build the RAG pipeline
- Pipeline documents included: Barry Boehm's Software Engineering and Software Engineering Economics, Cost Estimating Guide from Government Accountability Office, and NASA's Cost Estimating Handbook
- Language model used: meta-llama/Meta-Llama-3.1-8B-Instruct
 - Modular, can use other LLMs here



Creating the GUI

- The Python Gradio (<https://www.gradio.app/>) library was used to create a chatbot interface.

Cost Modeling Intelligent Chatbot

This chatbot references various cost modeling documents, such as NASA's Cost Estimating Guide and Software Engineering Economics.



NOTE: This chatbot is experimental, and all responses should be verified.

What would you like to know about cost modeling?

QUERY

Ask about Cost Modeling

Answer

RESPONSE

Document Section Reference

HISTORY



Demonstration



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Some of our other research:

- *“Leveraging Generative AI to Create, Modify, and Query MBSE Models”*
- *“Benchmarking Systems Engineering LLM Performance with SysEngBench”*
- *“Baselining AI Performance in Cost Modeling”*
- *“Systems Modeling with Large Language Models”*
- *“Systems and Software Engineering Cost Modeling of AI Assistance”*
- *“LLM Optimization for Systems Modeling”*



An aerial photograph of a large, historic university campus. The main building is a long, white, multi-story structure with a red-tiled roof and numerous windows. To the right of the main building is a large, rectangular, blue swimming pool. The campus is surrounded by lush green trees and a well-maintained lawn. In the background, a large, forested hill rises, dotted with houses and buildings. The sky is clear and blue.

Questions?